On the Effectiveness of Lipschitz-Driven Rehearsal in Continual Learning

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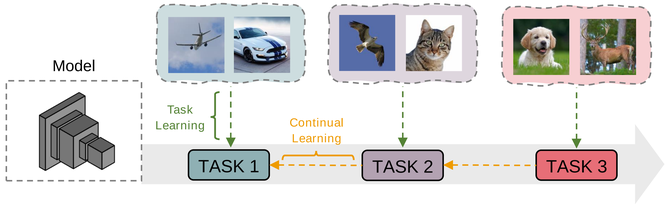
CONTINUAL LEARNING FRAMEWORK

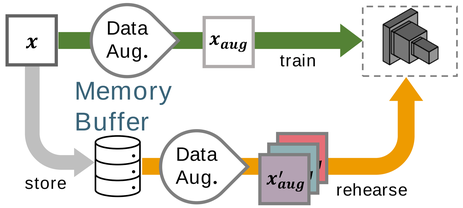
1. *Intro*: Continual Learning

Continual Learning (CL) seeks to enable learning from a stream of *tasks*, while avoiding *catastrophic forgetting*.



Rehearsal methods (*e.g.*, ER, ER-ACE, DER++, X-DER, iCaRL, GDumb, MER [6]) mitigate forgetting by *storing* and then *replaying* a subset of data from past tasks.

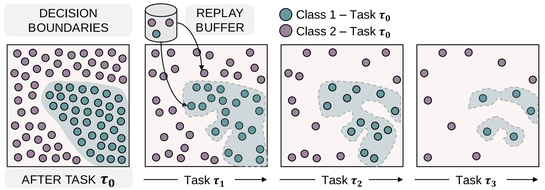
Problem: *repeated replay of few samples stored in the buffer leads a model to* **overfit** them!



2. *Problem*: Buffer Overfitting entails Bad Decision Boundaries

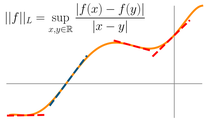
Consequence: *replay approaches becomes less and less effective at preventing forgetting.*

Model's **tolerance** to **input perturbations** as newer tasks incur, in the form of a decision surface plot.



Issue of rehearsal approaches: the area correctly predicted **tightens** and **degrades** around buffer data points: the model becomes **more sensitive** to input perturbations.

3. *Intuition*: Desired Effect on Boundaries

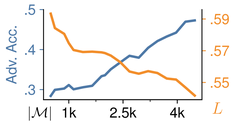


Lipschitz continuity. A function f is *Lipschitz continuous* if there exists a scalar L such that:

$$\|f(\mathbf{x}) - f(\mathbf{y})\|_2 \leq L\|\mathbf{x}-\mathbf{y}\|_2, \quad \forall \mathbf{x},\mathbf{y} \in \mathbb{R}^n.$$

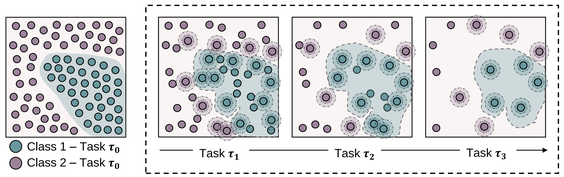
For a single $x \in \mathbb{R}^n$, the smallest L constitutes the **Lipschitz norm** $\|f\|_L$:

$$\|f\|_L = \sup_{x \neq y; y \in \mathbb{R}^n} \frac{\|f(\mathbf{x}) - f(\mathbf{y})\|_2}{\|\mathbf{x}-\mathbf{y}\|_2}.$$



Objective: *Restrict amount of change for buffer samples.*

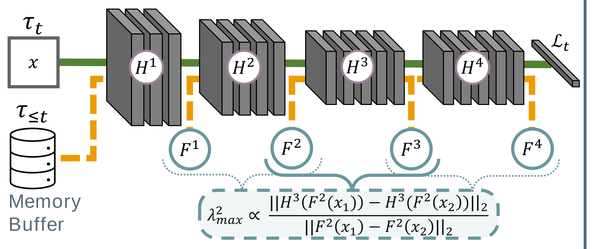
IDEA: ***push*** *the decision boundaries by encouraging the model to output similar predictions around* ***buffer datapoints***; *thus achieving* ***smoother*** *decision surfaces.*



4. Our approach

Lipschitz-DrivEn Regularization (LiDER)

We **constrain** the Lipschitz constants of several hidden layers but **only for buffer exemplars**.



Important: *we* ***do not*** *apply our regularization on the current task, as it already provides enough examples to avoid collapsing decision boundaries.*

5. Experimental Results

Final Avg. Accuracy [↑] and corresponding **gain** of LiDER on standard benchmarks.

Dataset	Split CIFAR-100		Split *mini*ImageNet		Split CUB-200	
Pre-train	✗		✗		*ImageNet*	
Buf. Size	500	2000	2000	5000	400	1000
iCaRL [5]	44.04 **+2.98**	50.23 **+0.98**	22.58 **+0.64**	22.78 **+1.17**	56.52 **+0.60**	60.09 **+0.28**
DER++ [2]	37.13 **+2.12**	52.08 **+1.19**	23.44 **+4.89**	30.43 **+4.61**	49.3 **+8.60**	61.42 **+6.55**
X-DER [1]	44.62 **+0.60**	54.44 **+0.27**	26.38 **+2.77**	29.91 **+2.65**	58.23 **+1.77**	64.9 **+1.08**
GDumb [4]	9.28 **+0.94**	19.69 **+6.46**	15.22 **+0.02**	27.79 **+1.70**	9.36 **+0.31**	18.98 **+0.53**
ER-ACE [3]	36.48 **+1.95**	48.41 **+1.91**	22.6 **+1.53**	27.92 **+2.08**	41.83 **+9.06**	51.98 **+8.94**

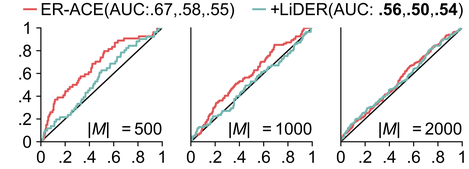
6. *Model Analysis*: Buffer Poisoning & Guessing Game

p	DER++	+LiDER
.0%	37.14	**39.25**
.01%	36.13	**38.08**
.1%	31.35	**35.53**
.25%	28.74	**30.78**

Buffer Poisoning. *How much do mislabeled buffer datapoints affect results of* ***rehearsal*** *approaches?*

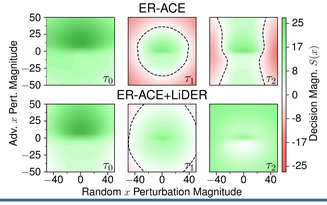
We observe a severe degradation; however, **LiDER** mitigates such an issue as it restrains the model from overfitting them.

Buffer Guessing Game. *Can we* ***identify*** *which examples were stored in the buffer just by looking at their corresponding* ***predictions***?



7. Effect on Boundaries

Decision boundaries for samples from the buffer become ***less strict*** *with* ***LiDER****.*



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